OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: TRINA SOLAR
MODEL / SERIES: Trina 310W (TSM-310DD05A.08)
DOCUMENT TITLE: Trina Solar Honey M Plus TSM-DD05A.08(II) Datasheet
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1. PRODUCT IDENTIFICATION & ARCHITECTURE

Cell Technology:	Polycrystalline
Module Construction:	Monofacial
Number of Cells / Format:	60 cells (6x10) multicrystalline / percium
Glass / Front Encapsulant:	3.2 mm High Transmission AR Coated Tempered Glass
Frame Material:	Anodized Aluminium Alloy (Black Frame .08)
Junction Box / Protection:	IP67 / IP68 with 3 bypass diodes

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)

Nominal Maximum Power (Pmax / Pnom):	310.0 Wp
Module Aperture Efficiency:	18.90 %
Open Circuit Voltage (Voc):	40.20 V
Short Circuit Current (Isc):	9.94 A
Maximum Power Voltage (Vmp):	33.10 V
Maximum Power Current (Imp):	9.37 A
Maximum System Voltage:	1000.0 V DC
Maximum Series Fuse Rating:	20.0 A

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS

Temp. Coefficient of Pmax (gamma):	-0.390 %/C
Temp. Coefficient of Voc (beta):	-0.290 %/C
Temp. Coefficient of Isc (alpha):	+0.050 %/C
Nominal Cell Temperature (NOCT/NMOT):	44 +/- 2 C (NOCT)
Operating Temperature Range:	-40 C to +85 C

4. MECHANICAL DIMENSIONS & RATINGS

Length:	1650.0 mm
Width:	992.0 mm
Thickness (Height):	35.0 mm
Gross Aperture Area:	1.6368 m2
Module Weight:	18.6 kg
Mechanical Design Load (Front):	5400 Pa (Snow load)
Mechanical Design Load (Rear):	2400 Pa (Wind load)

5. WARRANTY & LIFETIME PERFORMANCE

Product Workmanship Warranty:	10.0 Years
Linear Power Output Warranty:	25.0 Years
Annual Linear Degradation:	0.70 % / year
End of Warranty Retained Output:	80.7 %

6. PROVENANCE & ARCHIVAL METADATA

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SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730